CSR Spending & Impact Analysis Across Companies and States

1. Project Overview and Objective

This project aims to uncover insights from CSR spending data to understand allocation patterns and social impact focus areas. The main objectives of this project are:

- Analyze total CSR expenditure
- Identify top contributing companies
- Compare CSR spending across states
- Examine sector-wise CSR fund allocation
- Analyze year-wise CSR spending trends
- Identify underfunded social sectors

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2. Data Sources

- **Source Description and Timeline:** India Data Portal/ Ministry of corporate affairs/ Social Welfare **and** 2014-2020. https://indiadataportal.com/p/corporate-social-responsibility-csr/r/mca-csr_projects-dt-yr-iuo
- **Domain**: Social Welfare

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3. Problem Statement

- To analyse how companies allocate Corporate Social Responsibility (CSR) fund
- To evaluate which sector receives the maximum fund
- To analyse distribution of CSR spending across states
- Year-wise CSR trends
- To examine companies with highest CSR contribution

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4. Attribute (Column /Features) Details:

Raw Data:

Attribute Name	Data Type	Description
year	Date	Financial year(Jan 1st of every year)
state_name	categorical	Name of the state
state_code	Whole Number	Code of the state
district_name	string (Text)	Name of the district
district_code	Whole Number	Code of the district
csr_project_name	string (Text)	Name of the project
development_sector	Categorical	Social sector
company_name	string (Text)	Name of the company

cin	string (Text)	Unique identification for the company
company_class	string (Text)	Class of the company
company_type	string (Text)	Type of company
roc	string (Text)	Registrar of company
company_category	string (Text)	Category of company
listing_status	string (Text)	Listed/Unlisted
project_outlay_amount_in_lakhs	currency	Approved budget for the project
amount_spent_in_lakhs	currency	Actual amount utilized

CSR_final(Fact Table)

Attribute Name	Data Type	Description
year	Whole Number	Year
financial_year	Text	financial year
state_code	Whole Number	unique code for the state
csr_project_name	Text	name of the project
development_sector	Text	sector
standardized_sector	Text	grouped sector
company_name	Text	name of the company
company_type	Text	type of the company
Winsorized_Outlay_in_lakhs	Decimal Number	budget allocated
Winsorized_Spent_in_lakhs	Decimal Number	amount spent
winsorized_Spending_gap_in_lakhs	Decimal Number	remaining amount
Utilization%	Decimal Number	utilized percentage

state_dim(dimension table)

Attribute Name	Data Type	Description
state_name	Text	name of the state
state_code	Whole Number	unique code of the state

year_dim(dimension table)

Attribute Name	Data Type	Description
year	Whole Number	Year
financial_year	text	Financial year

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
    - Extracted the **Year** from the date column and created a new **Financial Year** column using:
      =YEAR(A2)-1 & "-" & YEAR(A2) to enable time-based analysis.
    - Removed non-analytical and irrelevant columns such as:
      id, district_code, district_name, cin, company_class, company_category, state_code, roc, listing_status
      to streamline the dataset for CSR-focused analysis.
    - Excluded rows containing **"Not Mentioned"** and **"NEC Not Mentioned"** in:
        1. state_name (not geographically meaningful)
        2. development_sector (not useful for sector-level analysis)
        3. csr_project_name (non-informative entries)
    - Removed records where both project_outlay_amount_in_lakhs and amount_spent_in_lakhs were **0**, as they had no financial impact on the analysis.
    - Standardized csr_project_name values by correcting inconsistencies (e.g., abbreviated forms expanded appropriately) to improve data quality and ensure accurate grouping.
    - Cleaned and trimmed state_name to remove extra spaces and formatting inconsistencies.
    - Standardized company_name by replacing:
        1. "Limited" → "Ltd"
        2. "Private Limited" → "Pvt Ltd"
           to ensure uniform naming conventions.
    - Removed duplicate records to maintain data integrity.
    - Created a new column **standardized_sector** using a mapping table to consolidate similar categories into 9 broader analytical groups via:
      =VLOOKUP([@[development_sector]], Sector_Mapping!A:B, 2, 0)
        - Standardized categories include:
            1. Education & Skill Development
            2. Health, Sanitation & Nutrition
            3. Environment & Sustainability
            4. Rural & Livelihood Development
            5. Women & Child Welfare
            6. Social Welfare & Inclusion
            7. Government & Relief Funds
            8. Art, Culture & Sports
            9. Others
    - Identified an anomalous **negative project_outlay_amount_in_lakhs** value. Since the corresponding spending was positive and no reversal entry existed, it was treated as a data entry error and corrected using =ABS(I2) to maintain financial consistency.
    - Removed negative values from amount_spent_in_lakhs as negative CSR expenditure is logically inconsistent.
    - Although extreme values are common in financial datasets, certain values were found to be unrealistically high. These were treated using the **Interquartile Range (IQR) method**, which captures the spread of the middle 50% of data, to identify and manage statistical outliers without compromising overall data integrity.

	Outlay	Outlay_va	spent	spent_values
Q1	3	3	2	2
Q3	72	72	50	50
IQR	69	69	48	48
Lower Bou	-100.5	-100.5	-70	-70
Upper Bou	175.5	175.5	122	122

Formula =IF(OR([@[project_outlay_amount_in_lakhs]]<Y5,[@[project_outlay_amount_in_lakhs]]>Y6),"Outlier","Normal")

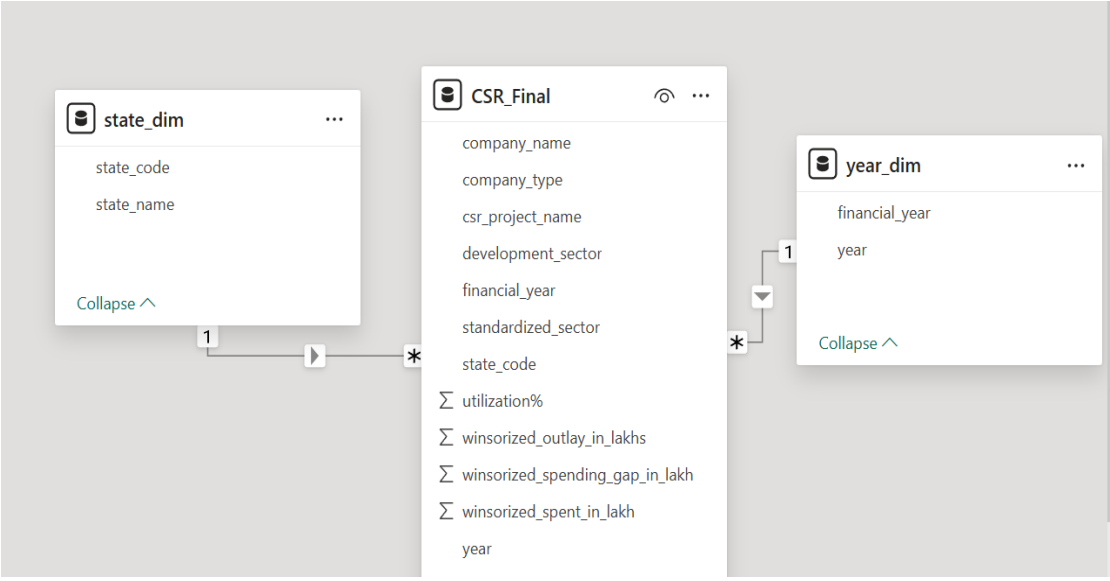
Q1	Formula=QUARTILE.INC(I:I,1)
Q3	Formula=QUARTILE.INC(I:I,3)
IQR	Formula = Q3-Q1
Lower Bou	Formula =Q1-1.5*IQR
Upper Bou	Formula =Q3+1.5*IQR

- Created calculated columns(winsorized_spend_gap, utilization%)
- Created a state_dim table and year_dim only with year

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

CSR_Final (state_code)	*—◀—1	state_dim (state_code)	Active
CSR_Final (year)	*—◀—1	year_dim (year)	Active



- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals**. (List Out)**

  1. Total Outlay = SUM(CSR_Final[winsorized_outlay_in_lakhs])
  2. Total Spent = SUM(CSR_Final[winsorized_spent_in_lakh])
  3. Total Gap = SUM(CSR_Final[winsorized_spending_gap_in_lakh])
  4. Utilization Rate = DIVIDE([Total Spent],[Total Outlay]) – Percentage
  5. Total Companies = DISTINCTCOUNT(CSR_Final[company_name])

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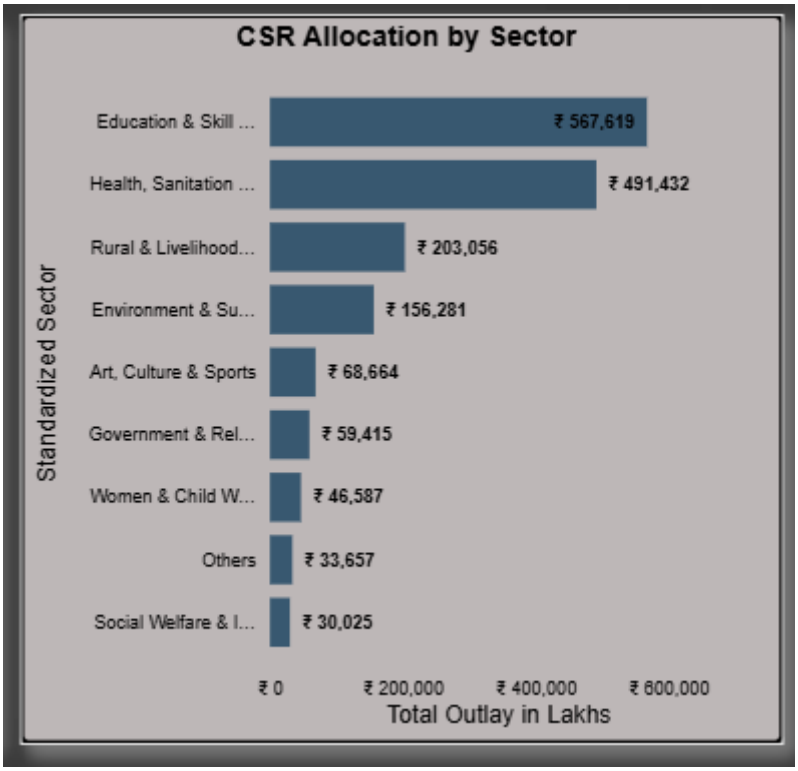
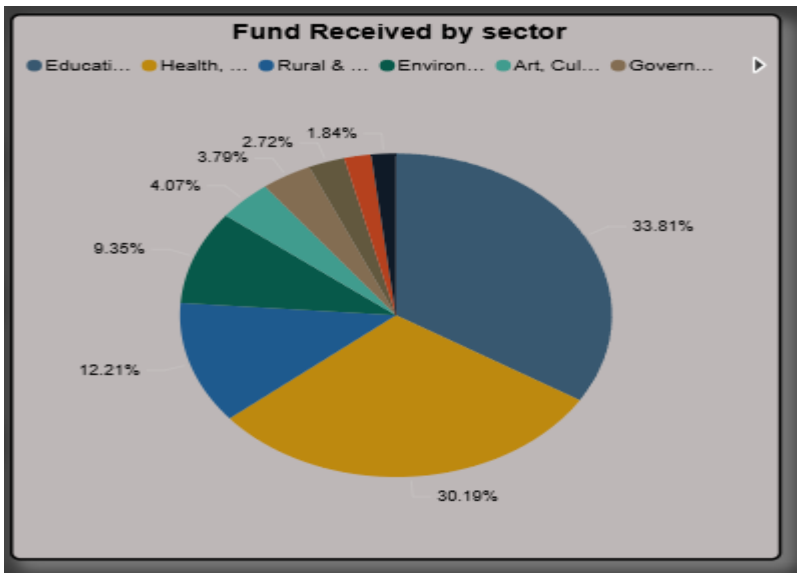
8. Analysis and Visualizations (Power BI)

Dashboard Features:

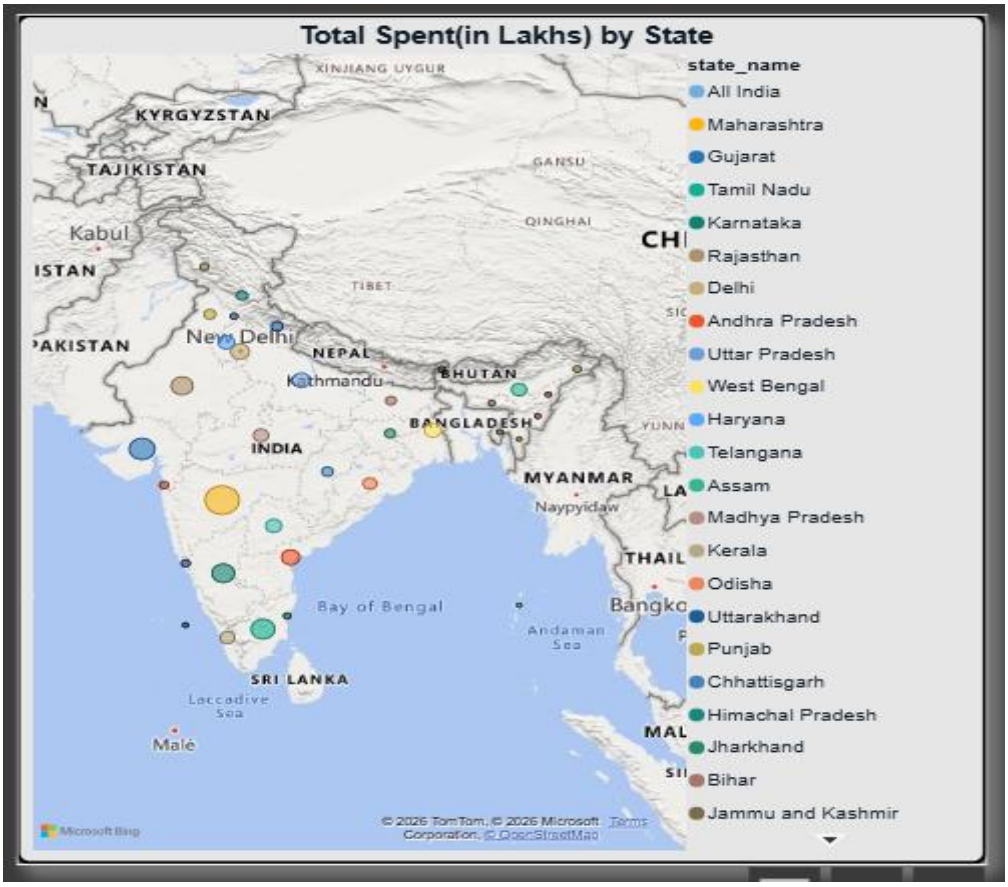
- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights. **(List Out)**



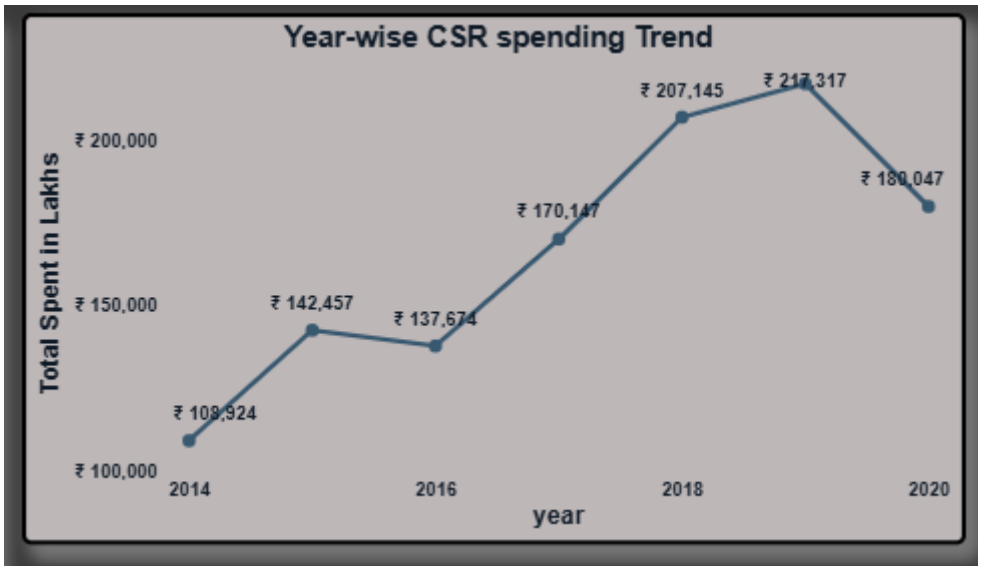
1. CSR allocation is heavily concentrated in Education (₹0.57M) and Health (₹0.49M), significantly outweighing other sectors.
2. This suggests that companies prioritize foundational development areas such as education and healthcare, while sectors like social inclusion and women welfare receive comparatively minimal funding.
3. CSR strategy appears focused on long-term human capital development rather than cultural or social equity programs.



1. **Education & Skill development** receives the highest CSR funding of 33.81%
2. Indicates that companies strongly prioritize **education, employability, and skill enhancement**.



1. The state-wise CSR spending analysis reveals that funds are concentrated in industrially developed states such as Maharashtra, Karnataka, and Tamil Nadu.
2. Less industrialized regions, particularly in the North-East, receive comparatively lower CSR funding, indicating a regional imbalance in CSR distribution.



1. The year-wise CSR spending trend shows a strong upward trajectory from 2014 to 2019, with spending nearly doubling over five years.

2. The peak was observed in 2019, indicating maximum corporate participation in CSR initiatives.
3. A decline in 2020 suggests possible economic disruptions; however, overall CSR commitment remains significantly higher compared to the initial years.

1. The CSR contribution analysis reveals that Infosys Ltd is the highest contributor among the top 10 companies, followed by Reliance Industries and Oil India Ltd.
2. A significant gap exists between the top contributor and the lower-ranked companies, indicating uneven CSR spending capacity across organizations.
3. Both private and public sector enterprises actively participate in CSR initiatives.

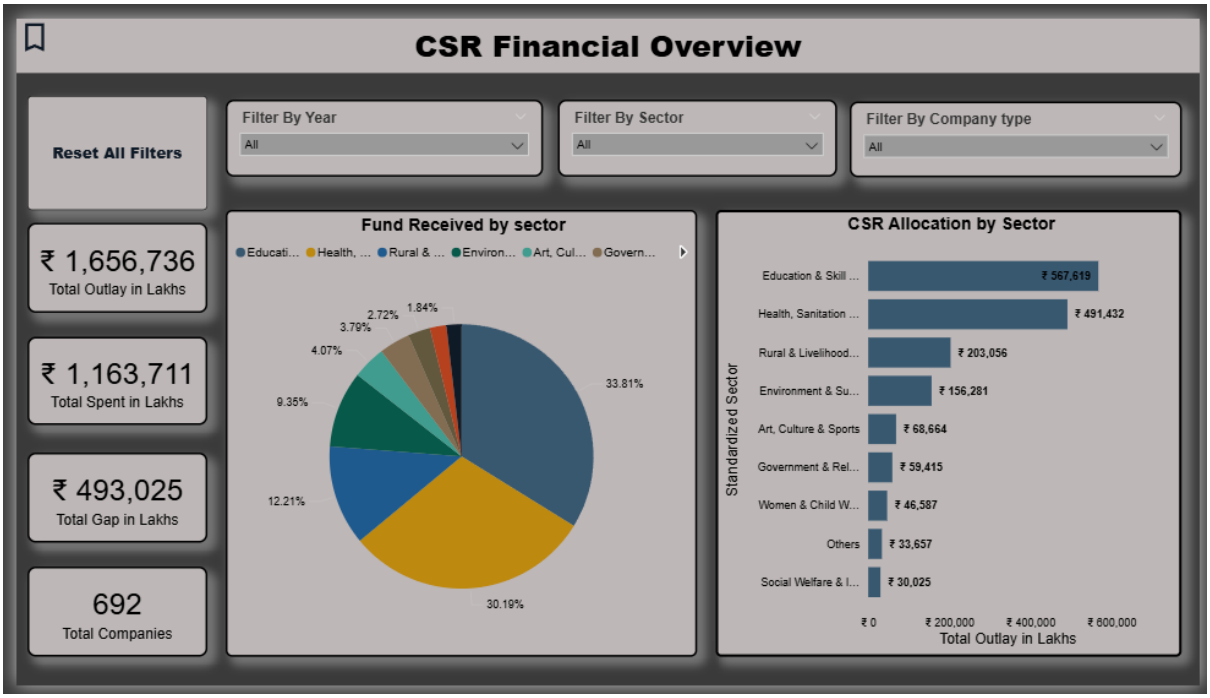
Year-wise CSR Spending by Sector

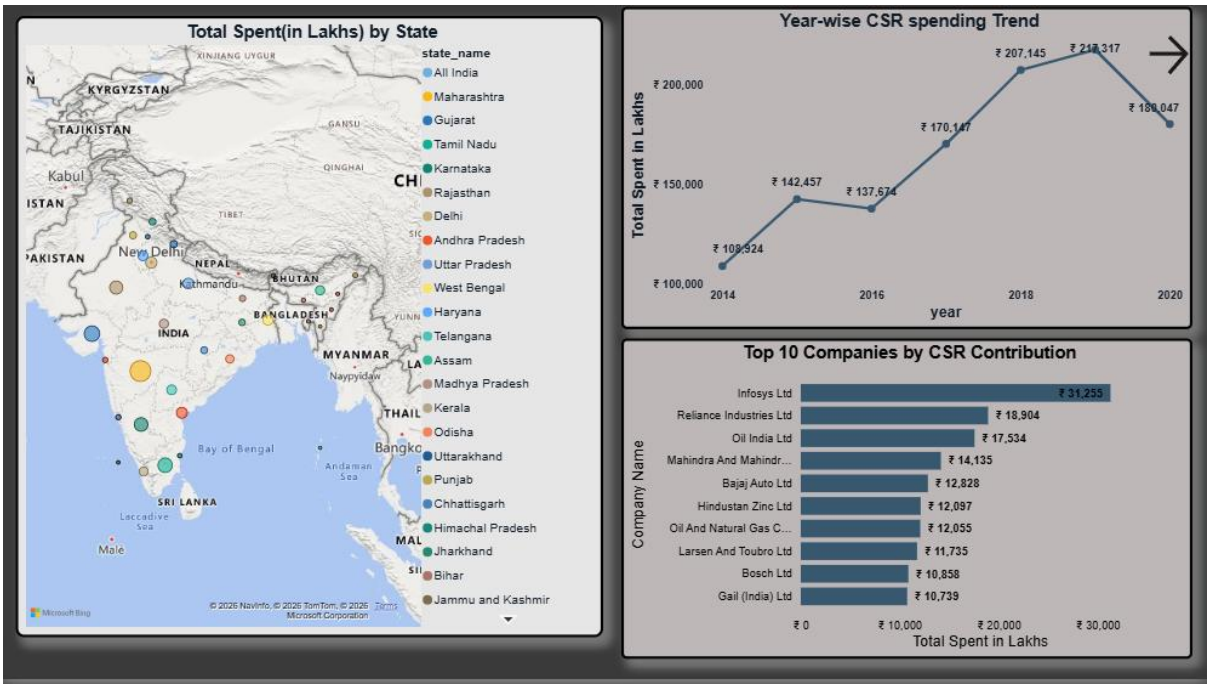
year	Total Outlay in Lakhs	Total Spent in Lakhs	Utilization Rate
2014	₹ 141,871	₹ 108,924	76.78%
2015	₹ 194,553	₹ 142,457	73.22%
2016	₹ 197,944	₹ 137,674	69.55%
2017	₹ 225,606	₹ 170,147	75.42%
2018	₹ 359,915	₹ 207,145	57.55%
2019	₹ 307,573	₹ 217,317	70.66%
2020	₹ 229,274	₹ 180,047	78.53%
Total	**₹ 1,656,736**	**₹ 1,163,711**	**70.24%**

1. Used Drill Down for better understanding of CSR.

2. Created a hierarchical matrix with Year and Sector to enable drill-down analysis. This allows users to evaluate sector-wise CSR spending within each year and compare trends across time.

SCREENSHOT OF DASHBOARD /REPORT





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Total	**₹ 1,656,736**	**₹ 1,163,711**	**70.24%**

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9. Insights & Conclusions

- **Key Findings:**
  1. Total CSR Outlay: ₹1,656,736 Lakhs
  2. Total CSR Spent: ₹1,163,711 Lakhs
  3. Overall Utilization Rate: 70.24%
  4. Education & Health together account for ~64% of total CSR funds
  5. CSR spending peaked in 2019
  6. 2020 shows a decline in spending
  7. CSR allocation is geographically concentrated in industrial states
  8. CSR contribution is dominated by a few large companies
- **Insights:**
  1. CSR funds are heavily concentrated in Education and Health, which together account for nearly two-thirds of total spending, while Social Welfare sectors receive the least allocation.
  2. CSR spending showed steady growth from 2014 to 2019, peaking in 2019, followed by a decline in 2020.
  3. The overall utilization rate is 70.24%, with the highest efficiency in 2020 and the lowest in 2018.
  4. Spending is concentrated in major industrial states and dominated by a few large companies, with Infosys being the top contributor.
  5. The low utilization in 2018 may be due to higher allocations combined with delayed or long-term project implementation.

6. The 2019 peak in spending reflects stronger CSR compliance and higher corporate profitability before the economic slowdown.
7. The 2020 decline and state-wise concentration are likely driven by pandemic-related disruptions and companies investing primarily in industrial regions where they operate.
8. CSR spending is likely to continue growing long-term.
9. Education & Health will remain dominant sectors.
10. Utilization rates may improve as monitoring becomes stronger.
11. Regional imbalance may continue unless policy incentives change.
12. Large corporations will continue dominating CSR contribution.
13. CSR allocation should be more diversified by increasing funding for under-supported sectors like Women & Child Welfare and Social Inclusion.
14. Improving monitoring mechanisms can help reduce the gap between allocated and actual spending, thereby increasing utilization efficiency.
15. Greater regional balance can be achieved by incentivizing CSR investments in underdeveloped and rural regions, including the North-East.
16. Encouraging participation from mid-sized companies and implementing strong impact measurement frameworks will strengthen overall CSR effectiveness.

10. Conclusions

The CSR analysis reveals that spending is heavily concentrated in education and healthcare sectors, with strong growth observed between 2014 and 2019. While overall CSR commitment has increased significantly over time, utilization efficiency and regional imbalance remain areas for improvement. Strategic diversification and improved fund monitoring can enhance the long-term impact of CSR initiatives.